

Nikhil Verma

Santa Clara, California 95050 • +1 9496782129 • Nikhilv1@uci.edu • Nikhil-Verma.me • [LinkedIn](#)

EDUCATION

UNIVERSITY OF CALIFORNIA, IRVINE

Master of Computer Science

CGPA: 3.91/4 • Coursework: Artificial Intelligence, Generative Models, Advanced Programming and Problem Solving, Operating Systems

Irvine, California

SEP 2023 - DEC 2024

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY

Electronics and Computer

CGPA: 3.68/4 • Coursework: Data Structures and Algorithms, Operating Systems, Cybersecurity

Patiala, India

JUN 2017 - JUN 2021

SKILLS

Backend & Systems: Distributed Systems, Microservices, Event-Driven Architectures, REST APIs, Observability (CloudWatch)

Languages & Frameworks: Java, Python, Golang, Spring Boot, TypeScript/JavaScript, Flask, Django

Cloud & Infrastructure: AWS (Lambda, API Gateway, DynamoDB, S3, Step Functions, EC2 & Image Builder, CloudWatch, CDK), Docker, Kubernetes, Terraform, CI/CD (Jenkins, GitLab)

Databases & Storage: DynamoDB, PostgreSQL, MongoDB, OpenSearch, ChromaDB

AI & LLMs: Multi-Agent Systems (LangGraph, LangChain, Strands), Amazon Bedrock AgentCore, Model Context Protocol (MCP), Agentic RAG, ReAct, Prompt Engineering, LLM Evaluation, Vector Databases (OpenSearch, ChromaDB), Amazon Bedrock, Claude

PROFESSIONAL EXPERIENCE

Amazon

Santa Clara, CA

Software Development Engineer I

FEB 2025 - Present

- Designed and shipped multi-product instance support for AWS License Manager User Subscriptions as sole low-level-design author, delivering 40+ changes across Java Lambda service logic, DynamoDB persistence, and EC2 Image Builder AMI pipelines. Multiple Microsoft products (Office, Visual Studio) now run on one EC2 instance, halving customer compute cost against the prior one-product-per-instance model.
- Designed and implemented a distributed cross-account user-subscription workflow in Java: modeled the CrossAccountRelationship entity, migrated a standalone DynamoDB table into a GSI, and built sync and async authorization across Lambda, Step Functions, and DynamoDB, lowering customer billing by \$10 per user per month.
- Led a security analysis of Visual Studio build components that showed a static, non-rotatable product key could be captured on-host (Windows Event ID 4688). Rejected 4 delivery designs and prevented a fleet-wide product-key leak.
- Onboarded the full User Subscriptions service (206 production accounts across 34 regions) to the AWS DevOps agentic-ops platform via CDK, integrating Ticketing, Medic, Rome, MWS, and Timber across 16 pipelines.
- Built Ticket Genie, an internal Python/AWS multi-agent tool (Amazon Bedrock/Claude) that classifies and triages roughly 800 tickets a month via Lambda and Step Functions, replacing manual triage and cutting time to resolution.
- Extended Java backend services and REST APIs to add Windows Server 2025 support and built the backend for Global Licensing Insights, an AWS re:Invent launch. Integrated License Manager data with Tagris for tag-based reporting and added CloudWatch metrics, alarms, dashboards, and canaries for cloud readiness (ORR).

Amazon

Santa Clara, CA

AI Engineer Intern

JUN 2024 - SEP 2024

- Built a multi-agent, dynamic code-generation pipeline that converts complex legal documents into machine-readable licensing statements, saving 100+ manual hours a month.
- Stack: LangGraph, LangChain, ReAct agents, ChromaDB and OpenSearch vector stores, and AWS Bedrock, orchestrated with internal MCP servers (built equivalent agent-tool capability from scratch, as MCP had not yet released).
- Designed a retrieval-augmented workflow with semantic chunking and similarity search, reaching 83% exact-statement accuracy in internal tests.

DEPARTMENT OF INFORMATICS, UNIVERSITY OF CALIFORNIA, IRVINE

Irvine, CA

Graduate Student Research Assistant

JAN 2024 - DEC 2024

- Led NSF-funded research on MLLMs under Dr. Anne Marie Piper, building and benchmarking an accessible version of ChatGPT (Next.js, MongoDB, Auth0, OpenAI APIs) to improve presentation accessibility for blind users.
- Engineered a custom Python module over Slack's API, providing an object-oriented abstraction that improved workplace communication accessibility for visually impaired users by 75%.

AMADEUS SOFTWARE LABS INDIA PVT LTD

Bangalore, India

Full Stack Developer - I

AUG 2021 - AUG 2023

- Owned the full lifecycle of an in-house data-driven automation project (Golang, Python, React) from concept to delivery, saving 3 hours of manual work a day, cutting processing time by about 70%, and fully automating bug-reporting workflows.

PUBLICATIONS & PATENTS

Patent (IDF filed, primary inventor): TABS (Token Accounting & Budget System), which extends License Manager to meter and enforce per-user token budgets for AI/LLM spend

Built a live end-to-end POC.

N. Verma, S. Kumari and P. Jain, "Post Quantum Digital Signature Change in IOTA to Reduce Latency in Internet of Vehicles (IoV) Environments," 2022 International Conference on IoT and Blockchain Technology. Improved IOTA encryption speed by 81% by replacing W-OTS with Dilithium for IoV networks.

PROJECT

XTC COIN

- Built a Python blockchain-style cryptocurrency backend: ledger API, multi-peer support, pytest suite, virtualenv tooling.